

Predictive Analytics Using Historical Data

This project demonstrates predictive analytics using historical sales data. A Linear Regression model was applied to forecast future sales trends. The workflow includes data preprocessing, model training, evaluation, and trend visualization.

Sample Historical Dataset

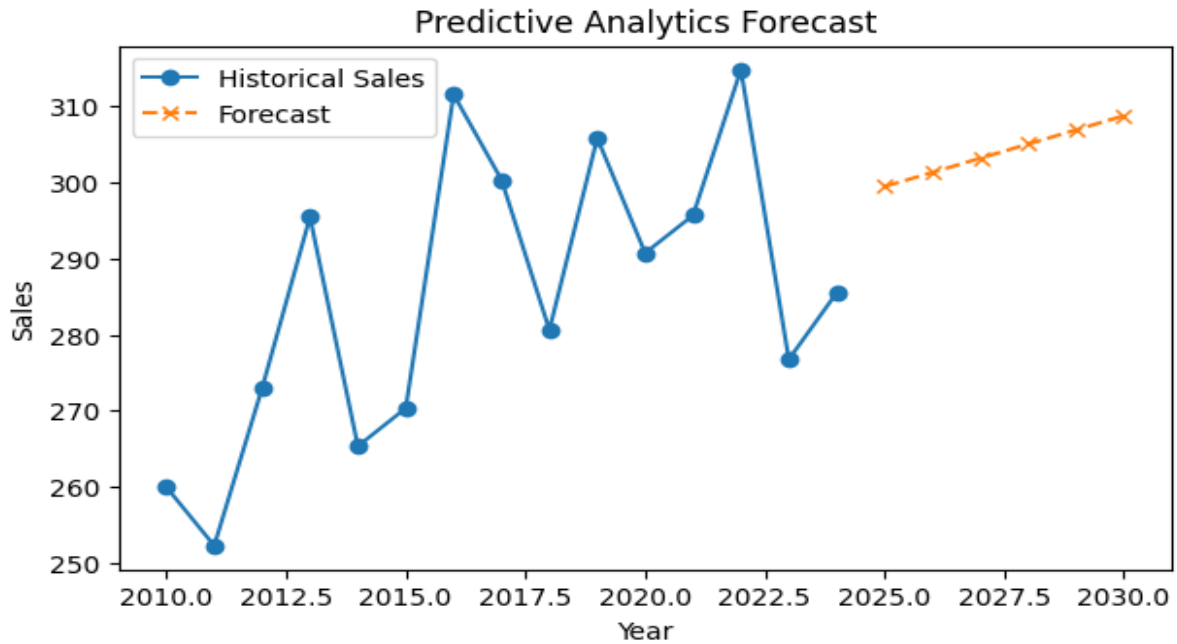
Year	Sales
2010.0	259.93428306022463
2011.0	252.2347139765763
2012.0	272.95377076201385
2013.0	295.4605971281605
2014.0	265.31693250553326
2015.0	270.3172608610164
2016.0	311.58425631014785
2017.0	300.3486945830582
2018.0	280.61051228130094
2019.0	305.8512008717193

Model Evaluation

Mean Absolute Error (MAE): 10.94

R² Score: 0.61

Trend Forecast Visualization



Future Forecast Results

Future Year	Predicted Sales
2025.0	299.47
2026.0	301.33
2027.0	303.2
2028.0	305.06
2029.0	306.92
2030.0	308.79

Conclusion

The predictive model successfully identified historical sales trends and forecasted future outcomes. This project highlights the importance of data cleaning, regression modeling, model evaluation, and visualization in predictive analytics workflows.